

domain-specific data. For example, Scholar GPT may be trained on academic papers, textbooks, scientific articles, research abstracts, and citation formats, thereby adjusting the model's weights to better handle scholarly language and knowledge.

Prompt engineering, on the other hand, does not alter the model's weights. Instead, it uses carefully crafted prompts or examples to guide its behaviour. This is why two custom GPTs—say, Monday and Scholar GPT—could be powered by the exact same underlying model, yet one may respond in a sarcastic tone while the other replies with professionalism and academic rigour.

The next step in enhancing a custom GPT is to combine the base GPT model with external retrieval systems. This is important because GPT models like GPT-3 or GPT-4 have a fixed knowledge cut-off date and cannot access live data. By integrating retrieval capabilities, a custom GPT can pull information from frequently updated external repositories, ensuring its responses remain current. For example, Scholar GPT can tap into academic databases like arXiv to provide more up-to-date answers.

Another way to empower a custom GPT is through the creation of a tailored interface. Many custom GPTs are deployed within specialised wrappers such as web apps, chatbots, or plugins. These interfaces can define the custom GPT's personality, tone, and interaction style. For instance, the Astrology Birth Chart GPT presents itself as an expert astrologer that requests your birth details to answer questions—a clear departure from the default ChatGPT interface.

Finally, custom GPTs can evolve further by incorporating user feedback and applying reinforcement learning—a machine learning approach that guides an AI to make better decisions through a reward system. This process helps the model better capture user intent within its domain and refine its outputs over time.

Now, let us explore some important custom GPTs by category. You can browse a wide variety of them by clicking the 'GPTs' option in the left sidebar of the ChatGPT interface. This section features both trending and popular custom GPTs, as well as those developed by OpenAI itself. The available categories include Writing, Productivity, Research & Analysis, Education, Lifestyle, DALL·E, and Programming.

First, let us look at some custom GPTs in the Research & Analysis category. Please note that the opinions expressed here are entirely personal. As an academic, my top choice in this category is Scholar GPT (its logo is shown in Figure 1), which we have already discussed earlier. To evaluate how it differs from the standard ChatGPT, I posed the same question to both: "Explain Grover's algorithm." Keep in mind that Grover's algorithm is a quantum search algorithm and can be challenging to grasp. After reviewing their responses, my conclusion is that ChatGPT's answer is straightforward and concise, whereas Scholar GPT not only offers intuitive insights but also preserves a strong sense of mathematical rigour. For this reason, I found Scholar GPT's explanation more satisfying.

Some other notable custom GPTs in the Research & Analysis category include: Consensus GPT

used for summarising research papers with citations, SciSpace GPT used for explaining academic papers in simpler language, Wolfram GPT used for advanced computation, data analysis, and scientific queries; AI Lawyer GPT used for summarising laws, explaining legal concepts, and generating legal drafts; and Marketing GPT used for creating advertisements, planning campaigns, and analysing target audiences.

Now, let us turn to the Education category of custom GPTs. My top pick here is Math GPT (stylised as math), which offers three distinct modes of operation: Tutor/Problem Solving mode, Graph mode, and Calculator mode. The Tutor/Problem Solving mode provides step-by-step explanations, guided learning, and interactive problem-solving, making it ideal for students seeking conceptual clarity. The Graph mode is designed for plotting mathematical functions, visualising data, and generating custom graphs. For example, Figure 2 shows the output generated by Math GPT when given the prompt $y = \sin(x)$. The Calculator mode handles quick arithmetic, symbolic computation, and equation solving, making it a convenient tool for both simple and advanced calculations.

Some other notable custom GPTs in the category Education include: Academic Assistant Pro GPT used for organising research material, generating academic content, and providing structured writing assistance; Finance (Business Finance) GPT used for financial analysis, budgeting, forecasting, and generating business reports; AI for Medical Students GPT used for explaining medical concepts, summarising clinical guidelines, and supporting exam preparation; Chemistry Chem GPT used for solving chemistry problems, explaining chemical concepts, and generating molecular structures; and IELTS Speaking – English & Language Learning GPT used for practising IELTS speaking tasks,



Figure 1: The logo of Scholar GPT